
Parameter Scope Matters: MNE-L2 for Reliable Low-Latency ANN-to-SNN Conversion

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Abstract

1 Spiking neural networks (SNNs) perform event-driven computation with sparse
2 spike activity, making them promising for efficient edge inference. ANN-to-SNN
3 conversion provides a practical route to deep SNNs by transferring well-trained
4 ANN parameters, but the resulting discrete representations can be fragile under
5 noise. We identify a parameter-scope failure in regularization: applying L2 to batch-
6 normalization (BN) affine parameters can make converted SNNs substantially less
7 robust than applying it only to weights. We explain this behavior through a local
8 boundary-crossing analysis, in which representation changes are governed by the
9 effective perturbation relative to the quantization margin. Guided by this analysis,
10 we propose margin-normalized effective L2 (MNE-L2), which penalizes BN-folded
11 weight amplification relative to that margin while stopping the regularization
12 gradient through BN affine parameters and IF thresholds. Across five-seed CIFAR
13 experiments, MNE-L2 improves high-noise accuracy over all-parameter L2 by
14 up to 62.06 percentage points while reducing clean firing activity and estimated
15 arithmetic energy. These results establish regularization scope as an important
16 design choice for reliable low-latency SNN deployment.

17 1 Introduction

18 Spiking neural networks (SNNs) communicate through discrete spikes and support sparse, event-
19 driven computation, offering a promising route to efficient inference on neuromorphic hardware [1].
20 High-accuracy deep SNNs can be obtained either by direct training or by converting artificial neural
21 networks (ANNs). ANN-to-SNN conversion is particularly practical because it retains conventional
22 ANN optimization and maps learned activations to spike-rate representations [2, 3]. Converted
23 SNNs, however, often require many simulation steps to approximate ANN activations. Reducing
24 the timestep budget lowers latency and event cost but increases finite-time conversion error [4].
25 Quantized activation training and related calibration methods have substantially improved accuracy at
26 low timesteps [5, 4], making the reliability of these discrete representations under deployment noise
27 increasingly important.

28 Unlike continuous ANN activations, SNN representations are formed by thresholded spike counts and
29 can change abruptly when perturbations alter firing events. Prior work has studied noise sensitivity
30 in trained SNNs and adversarial robustness after conversion [6, 7]. Most low-latency conversion
31 methods, however, focus on reducing clean conversion error through activation design, calibration, or
32 membrane-potential correction. The effect of the source ANN’s regularization scope on the robustness
33 of the converted SNN has received much less attention.

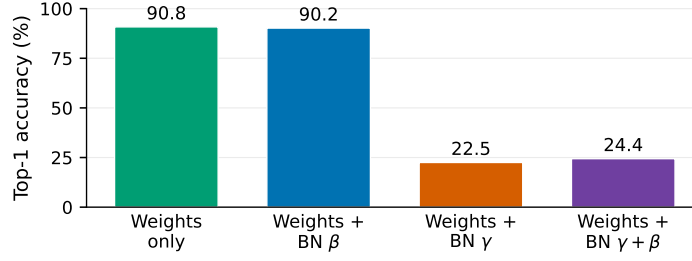


Figure 1: CIFAR-10 VGG16 parameter-scope ablation under noise at $L = T = 16$. Accuracy is shown at $\sigma = 1$ for a single seed-42 checkpoint per setting. Including BN- γ , with or without BN- β , causes the early collapse.

We uncover a parameter-scope failure in this regularization. In networks with batch normalization (BN) [8], the label “L2 regularization” is ambiguous unless the penalized parameter set is specified. We therefore compare weights-only L2 with selective penalties on weights and the BN affine parameters. As Figure 1 shows, including the BN gain γ causes an early accuracy collapse under noise, while weights-only L2 and weights plus BN bias β remain stable. This controlled observation is the starting point of our stability analysis.

To explain the failure, we analyze a threshold-quantized converted layer under additive perturbations. A discrete representation changes only when a perturbation crosses a quantization boundary, so local risk is governed by effective noise relative to the nearest margin. Downstream BN-folded weights determine how strongly perturbations are amplified, while IF thresholds determine the conversion margin. Guided by this analysis, we propose margin-normalized effective L2 (MNE-L2), which penalizes BN-folded amplification relative to that margin without directly updating BN affine parameters or IF thresholds through the regularization term.

Our contributions are:

- We identify a previously overlooked L2 parameter-scope failure and isolate BN- γ as the affine term that distinguishes the unstable settings in a controlled four-way ablation.
- We derive a local boundary-crossing view and layerwise margin-to-noise diagnostics that connect effective amplification to changes in discrete SNN representations.
- We formulate MNE-L2 and evaluate it against L1/L2 baselines using five-seed CIFAR VGG13/16/19 experiments, a preliminary ImageNet ResNet-18 extension, and full-test-set firing-density and arithmetic-energy estimates.

2 Related Work

SNN training and ANN-to-SNN conversion. High-performance SNNs are commonly obtained either by direct training or conversion. Direct approaches optimize temporal spiking dynamics through surrogate gradients [9–11], but require backpropagation through multiple simulation steps. Conversion instead trains a source ANN and maps its activations to firing rates. Early methods use weight normalization, threshold balancing, and activation normalization [12, 2]; later work scales conversion to VGG and residual architectures [3]. Our work follows conversion and targets the reliability of intermediate quantized representations at low T , rather than proposing a new neuron model or direct-training rule.

Quantization-based low-latency conversion. Finite-time mismatch has been reduced through potential initialization [13], parameter calibration [14], residual membrane-potential correction [15], offset-spike calibration [16], probabilistic temporal alignment [17], and quantized ANN training [5, 4]. Recent methods also learn conversion-error compensation or bridge quantized activations and spiking neurons [18–20]. These approaches mainly modify conversion dynamics, calibration, or activation design. MNE-L2 is complementary: the conversion rule is unchanged, while the source training objective penalizes layerwise amplification relative to the quantization margin. The BN-scope

71 ablation further shows that the parameter set receiving a nominally standard L2 penalty can materially
 72 change measured post-conversion reliability.

73 **Stability and noise sensitivity.** Perturbation stability has been analyzed through Lyapunov bounds
 74 for ANNs [21], membrane-potential dynamics in trained SNNs [6], spectral properties of random
 75 LIF networks [22], and classifier-level adversarial robustness after conversion [7]. We study a
 76 different target: whether a controlled post-input-IF perturbation changes intermediate QCFS codes
 77 and how these changes propagate through a converted network. The margin-to-noise and bin-crossing
 78 diagnostics therefore localize representation failures rather than certifying worst-case classifier
 79 robustness.

80 3 From Parameter Scope to Margin-Aware Regularization

81 3.1 Low-latency QCFS conversion

82 For activation x , threshold λ , and quantization level L , the source ANN uses

$$q_L(x; \lambda) = \frac{\lambda}{L} \left\lfloor L \operatorname{clip}\left(\frac{x}{\lambda}, 0, 1\right) + \frac{1}{2} \right\rfloor. \quad (1)$$

83 At inference, an integrate-and-fire (IF) layer emits a length- T spike sequence whose rate represents
 84 the quantized activation. Unless stated otherwise, we use $L = T = 16$ and a rate-uniform schedule
 85 for the VGG experiments. Additional depth-transfer and compact low-latency controls are reported
 86 only in the appendix.

87 3.2 Local noise-to-margin analysis

88 Let z be the clean output of an IF layer and $\tilde{z} = z + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Let $\mathcal{B}_l$ contain the QCFS
 89 boundaries of layer l and $d_l(u) = \min_{b \in \mathcal{B}_l} |u - b|$. Under a local additive model $\delta_l \sim \mathcal{N}(0, \sigma_{\text{eff},l}^2)$,
 90 changing the quantization bin requires crossing a boundary, so

$$\Pr[q_L(u + \delta_l) \neq q_L(u) \mid u] \leq 2 \mathcal{Q}\left(\frac{d_l(u)}{\sigma_{\text{eff},l}}\right), \quad (2)$$

91 where $\mathcal{Q}$ is the Gaussian upper-tail function. We estimate $\sigma_{\text{eff},l} = \sqrt{\mathbb{E}[(z_l - \tilde{z}_l)^2]}$, summarize $d_l(u)$
 92 by its median $d_{50,l}$, and report

$$\rho_l = \frac{d_{50,l}}{\sigma_{\text{eff},l} + \varepsilon}, \quad P_{\text{cross},l} = \Pr[q_L(z_l) \neq q_L(\tilde{z}_l)]. \quad (3)$$

93 Small ρ_l and large $P_{\text{cross},l}$ identify layers where noise is likely to change discrete spike counts.
 94 In an idealized independent aggregation model, σ_{eff}^2 is upper-bounded by $O(1/(CT))$, while the
 95 nominal QCFS half-cell margin decreases as $1/L$; hence a nominal stability proxy scales as $\sqrt{CT}/L$
 96 (Appendix A). This motivates evaluating precision, width, and timestep jointly rather than treating L
 97 as an isolated accuracy knob. It also suggests controlling the BN-folded amplification that contributes
 98 to $\sigma_{\text{eff},l}$ relative to the following QCFS margin.

99 3.3 MNE-L2 derived from the stability analysis

100 Guided by the preceding analysis, for each convolution or linear layer followed by BN and an IF
 101 activation, we fold the BN scale into the matched weight while stopping the regularization gradient
 102 through BN and the IF threshold:

$$\widetilde{W}_{l,o} = \operatorname{sg}\left(\frac{\gamma_{l,o}}{\sqrt{v_{l,o} + \epsilon_{\text{BN}}}}\right) W_{l,o}, \quad M_{\text{eff},l} = \frac{1}{C_l} \sum_o \|\widetilde{W}_{l,o}\|_F^2. \quad (4)$$

103 Margin-normalized effective L2 is

$$\mathcal{R}_{\text{MNE}} = \sum_l \frac{L_l^2 M_{\text{eff},l}}{\operatorname{sg}(\lambda_l)^2 + \varepsilon}, \quad (5)$$

where sg denotes stop-gradient. Layers without a following IF threshold are excluded. The objective uses the current folded scale to weight each matched W_l , but the penalty does not directly shrink $\text{BN-}\gamma$, $\text{BN-}\beta$, or λ_l . This distinction motivates the scope ablation: standard all-parameter L2, weights-only L2, and selective L2 on weights plus BN affine terms are different interventions even when their coefficients are numerically similar.

4 Experiments

4.1 Setup

We use CIFAR-10 and CIFAR-100 [23] with QCFS VGG16 and ImageNet [24] with QCFS ResNet-18 [25]. Unless stated otherwise, $L = T = 16$ and Gaussian noise is injected after the input IF. We denote optimizer L2 applied to all trainable parameters as L2-all. L2-wo restricts the same penalty to convolutional and linear weight tensors, excluding biases, BN affine parameters, and trainable IF thresholds; L1-wo uses the same weights-only scope. The parameter-scope ablation uses CIFAR-10 seed 42 and one stored perturbation stream at each $\sigma \in [0, 5]$. It compares weights-only L2, weights plus $\text{BN-}\beta$, weights plus $\text{BN-}\gamma$, and weights plus both affine terms. The cross-dataset CIFAR study evaluates five independently trained checkpoints (seeds 40–44) for L2-all, L2-wo, L1-wo, and MNE-L2. Curves show means and sample standard deviations across training seeds. Appendix C.2 repeats the CIFAR sweep with VGG13 and VGG19 and includes a fixed- α Calibrated MNE control.

The scale-transfer experiment evaluates the same four methods on one ImageNet ResNet-18 checkpoint (seed 42) over $\sigma \in [0, 2]$. Thus, the ImageNet result is a single-checkpoint extension, not a variance-aware benchmark. Hyperparameters are fixed before the noise evaluation, and the reported configurations are not clean-accuracy matched. The injected perturbation is a synthetic internal-noise stress test rather than an adversarial input attack [7] or a substitute for device measurement. CIFAR activity and energy accounting uses all 10,000 test images for each of five seeds at $T \in \{4, 8, 16\}$; only clean activity is used for deployment energy. Layerwise diagnostics use held-out CIFAR-10 examples. Further details are in the appendix.

4.2 BN gain distinguishes the unstable scope settings

Figure 1 isolates the BN affine terms. At $\sigma = 1$, weights-only L2 retains 90.80% accuracy and adding $\text{BN-}\beta$ retains 90.19%. Adding $\text{BN-}\gamma$ instead falls to 22.50%; adding both affine terms reaches 24.41%. At $\sigma = 3$, the first two configurations still achieve 75.63% and 70.65%, while both configurations containing $\text{BN-}\gamma$ are at chance. The near-overlap of the $\text{BN-}\gamma$ and $\text{BN-}\gamma + \beta$ curves implicates gain, rather than bias, as the principal scope difference in this controlled run.

The scope result is consistent with the layerwise conversion diagnostics. In late VGG blocks, all-parameter L2 and weights-plus- $\text{BN-}\gamma$ reach $\rho_l \approx 0.05\text{--}0.11$ and $P_{\text{cross},l} \approx 0.62\text{--}0.74$, whereas MNE-L2 and weights-only L2 retain larger margins and mostly stay below 0.15 crossings (Appendix B). This does not establish a universal BN mechanism: the four-way intervention uses one training seed. It shows that reporting “L2” without its parameter scope can conceal experimentally important variation.

4.3 MNE-L2 across CIFAR and ImageNet

Figure 2(a,b) extends the stress test to five independently trained checkpoints per CIFAR dataset. At $\sigma = 3$, MNE-L2 reaches 86.59% on CIFAR-10 and 55.19% on CIFAR-100, compared with 82.01% and 45.33% for weights-only L1. At $\sigma = 5$, the gap widens to 72.06% versus 39.40% on CIFAR-10 and 41.14% versus 17.91% on CIFAR-100. All-parameter L2 collapses near $\sigma = 1$ on both datasets, consistent with the scope ablation.

The high-noise ordering persists for VGG13 and VGG19 across both CIFAR datasets (Appendix C.2), indicating that it is not specific to VGG16 depth.

The single-checkpoint ImageNet result in Figure 2(c) follows the CIFAR ordering at moderate noise: MNE-L2 retains 33.07% at $\sigma = 1$, versus 21.11% for weights-only L1, and remains highest at $\sigma = 2$. This residual, larger-dataset extension is preliminary because it uses one seed.

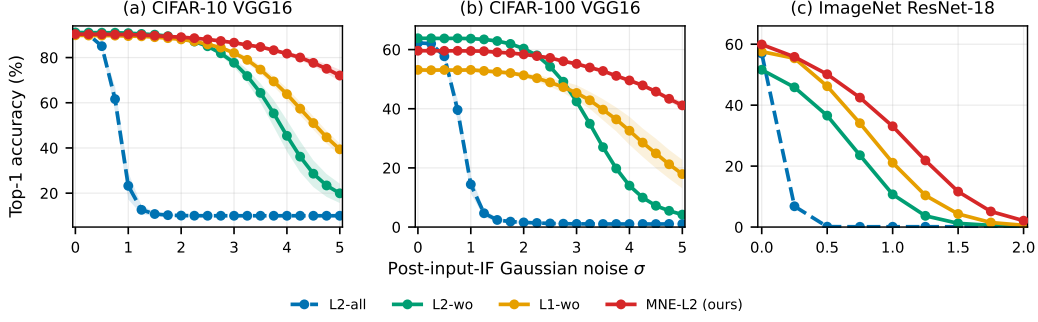


Figure 2: Post-input-IF Gaussian-noise sweeps at $L = T = 16$. (a,b) CIFAR VGG16 curves are five-seed means; bands show sample standard deviations across seeds 40–44. (c) ImageNet ResNet-18 uses one seed-42 checkpoint and therefore has no uncertainty band. MNE-L2 is highlighted in red and all data points use circular markers.

Table 1: Five-seed CIFAR VGG16 robustness–activity summary at $T = 16$ (mean $\pm$ sample standard deviation; 10,000 test images per seed). Accuracy is measured under post-input-IF Gaussian noise at $\sigma = 5$; firing density and arithmetic energy use clean inference. The best value in each dataset and metric is bold, and the second-best is underlined. Clean accuracy and full activity distributions appear in Appendix C.1.

Dataset	Method	Clean firing density	Clean energy (mJ)	Acc. at $\sigma = 5$ (%)
CIFAR-10	L2-all	0.105 ± 0.003	0.585 ± 0.006	10.00 ± 0.00
	L2-wo	0.102 ± 0.001	0.505 ± 0.003	19.92 ± 4.29
	L1-wo	0.077 ± 0.001	0.402 ± 0.005	39.40 ± 2.60
	MNE-L2 (ours)	<u>0.083 ± 0.001</u>	<u>0.472 ± 0.003</u>	72.06 ± 2.57
CIFAR-100	L2-all	0.106 ± 0.003	0.558 ± 0.008	1.05 ± 0.06
	L2-wo	0.105 ± 0.001	0.522 ± 0.002	4.21 ± 0.24
	L1-wo	0.084 ± 0.002	0.429 ± 0.005	17.91 ± 4.84
	MNE-L2 (ours)	<u>0.090 ± 0.001</u>	<u>0.494 ± 0.004</u>	41.14 ± 1.23

153 4.4 Clean sparsity, robustness, and estimated energy

154 Table 1 compares all four regularizers at high noise. On CIFAR-10, MNE-L2 lowers clean firing
155 from 0.105 to 0.083 and energy from 0.585 to 0.472 mJ relative to L2-all, while retaining 72.06%
156 rather than 10.00% accuracy at $\sigma = 5$. The same robustness ordering holds on CIFAR-100, where
157 MNE-L2 reaches 41.14% and L2-all reaches 1.05%. L1-wo is the sparsest and least energy-intensive
158 method, but Appendix C.1 shows that it trails MNE-L2 by 6.49 clean CIFAR-100 points.

159 Following Horowitz [26], we charge repeated first-layer MACs at 4.6 pJ and measured event-triggered
160 ACs in later layers at 0.9 pJ (Appendix Eq. 9). Only clean activity enters energy: CIFAR-10 L2-all
161 firing at $T = 4$ rises from 0.105 to 0.139 under $\sigma = 1$, so noisy activity would confound injected
162 events with deployment cost. Appendix C.1 shows that L2-wo is robust but nearly as active as L2-all;
163 L1 is sparsest, but trails MNE-L2 by 6.49 clean CIFAR-100 points. These estimates omit memory,
164 neuron updates, control, interconnect, and leakage, and are not device measurements.

165 5 Implications, Limitations, and Conclusion

166 For deployment, regularizer names should specify their parameter scope: including BN- γ changes
167 the observed robustness ranking. MNE-L2 instead weights convolutional and linear parameters
168 using folded BN statistics without shrinking affine or threshold terms, yielding lower activity than
169 L2-all and an intermediate point between sparse but inaccurate L1 and active, clean-accurate L2-wo.
170 Limitations include a one-seed scope study, one ImageNet checkpoint, synthetic single-site noise,
171 unmatched clean accuracy, and arithmetic-only energy; five-seed CIFAR evidence remains restricted
172 to VGG-family architectures. These constraints preclude universal dominance, but within the tested
173 settings MNE-L2 better preserves post-IF representations than L2-all.

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A A Local Precision–Width–Time Model

Consider an idealized normalized aggregation of independent perturbations,

$$\delta_{C,T} = \frac{1}{CT} \sum_{t=1}^T \sum_{i=1}^C w_i \epsilon_{i,t}, \quad \epsilon_{i,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (6)$$

If $C^{-1} \sum_i w_i^2 \leq M$, then

$$\text{Var}(\delta_{C,T}) = \frac{\sigma^2}{C^2 T} \sum_i w_i^2 \leq \frac{\sigma^2 M}{CT}. \quad (7)$$

For an unclipped QCFS cell, the nominal half-cell margin is $h_L = \lambda/(2L)$. Combining this value with the variance bound gives the nominal proxy

$$\rho_{\text{nom}} = \frac{h_L}{\sigma_{\text{eff}}} \geq \frac{\lambda \sqrt{CT}}{2L \sigma \sqrt{M}}. \quad (8)$$

Table 2: Five-seed clean MNIST conversion at $T = 4$ (mean $\pm$ standard deviation, no injected noise). Δ is $L=16$ minus $L=4$ SNN accuracy.

Channels	SNN, $L = 4$	SNN, $L = 16$	Δ
2/4	96.14 ± 0.91	87.80 ± 6.94	-8.34
4/8	98.00 ± 0.26	97.01 ± 0.60	-0.99
8/16	98.67 ± 0.13	98.22 ± 0.37	-0.44
16/32	98.67 ± 0.12	98.42 ± 0.21	-0.25

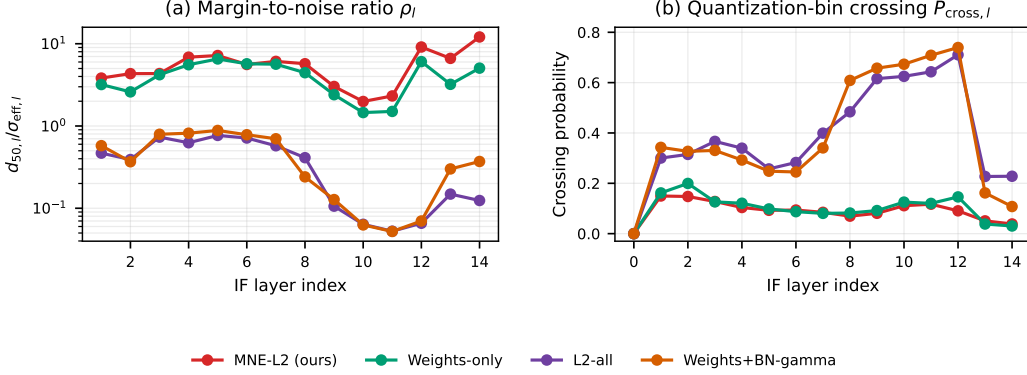


Figure 3: Layerwise CIFAR-10 VGG16 diagnostics at fixed $\sigma = 1$. All-parameter L2 and weights+BN- γ checkpoints develop small margin-to-noise ratios and high bin-crossing probability in later blocks.

Equation (2) follows because a changed quantization code requires $|\delta_l| \geq d_l(u)$ and a zero-mean Gaussian satisfies $\Pr(|\delta_l| \geq d) = 2Q(d/\sigma_{\text{eff},l})$. This local model isolates the competing directions of C , T , and L ; it is not a claim that all network perturbations are independent or Gaussian, nor a proof that the empirical accuracy gap must be monotone.

A.1 Compact precision-width control

To separate conversion error from injected-noise calibration, we train weight-decayed MNIST CNNs with channel pairs (2, 4), (4, 8), (8, 16), and (16, 32) at $L \in \{4, 16\}$. Each setting uses five training seeds and is converted at $T = 4$ with no injected noise. Raising L from 4 to 16 improves every source ANN by 0.06–0.29 points, yet reduces $T = 4$ SNN accuracy by 8.34 points for the 2/4-channel model. This loss contracts to 0.25 points at 16/32 channels (Table 2).

B Layerwise Diagnostics

Figure 3 reports the two diagnostics from Eq. (3). Layer 0 is the perturbed input IF; its effective-noise denominator is zero by construction for the downstream-propagation statistic and is omitted from the left panel. The right panel directly measures whether clean and noisy IF outputs differ.

C Additional Experimental Details

The CIFAR VGG16 source ANNs are trained for 300 epochs with batch size 128 and seeds 40–44 where multi-seed results are reported. Training uses SGD with momentum 0.9, initial learning rate 0.1, and cosine annealing. Random crops with four-pixel padding, horizontal flips, CIFAR AutoAugment, and Cutout are used during training; evaluation uses only dataset-specific normalization. MNE-L2 uses coefficient 10^{-4} with detached BN affine/statistical factors and detached IF thresholds. All-parameter L2 uses weight decay 5×10^{-4} , weights-only L1 uses 10^{-5} , and the selective L2 scope controls use coefficient 2.5×10^{-4} . The scope comparison reuses CIFAR-10 seed-42 checkpoints and the same evaluation protocol at every severity.

Table 3: Top-1 accuracy under post-input-IF noise at $\sigma = 1$ (mean $\pm$ sample standard deviation across seeds 40–44).

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
CIFAR-10	L2-all	10.10 ± 0.15	11.17 ± 1.18	23.30 ± 6.46
	MNE-L2 (ours)	88.97 ± 0.21	89.76 ± 0.18	90.09 ± 0.10
CIFAR-100	L2-all	1.56 ± 0.40	2.88 ± 0.83	14.64 ± 3.16
	MNE-L2 (ours)	58.77 ± 0.28	59.27 ± 0.11	59.48 ± 0.23

Table 4: Full clean SNN top-1 accuracy (mean $\pm$ sample standard deviation across seeds 40–44).

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
CIFAR-10	L2-all	90.29 ± 0.23	90.52 ± 0.14	90.59 ± 0.13
	L2-wo	90.57 ± 0.18	90.96 ± 0.06	91.10 ± 0.07
	L1-wo	89.09 ± 0.25	89.70 ± 0.30	89.87 ± 0.20
	MNE-L2 (ours)	89.93 ± 0.16	90.31 ± 0.17	90.38 ± 0.14
CIFAR-100	L2-all	61.83 ± 0.47	62.22 ± 0.49	62.27 ± 0.37
	L2-wo	62.68 ± 0.54	63.57 ± 0.31	63.83 ± 0.26
	L1-wo	52.37 ± 0.95	52.89 ± 0.71	53.13 ± 0.73
	MNE-L2 (ours)	59.24 ± 0.21	59.62 ± 0.31	59.62 ± 0.22

285 The ImageNet ResNet-18 source ANNs are trained at $T = 0$ for 90 epochs with seed 42, batch size
 286 256, SGD momentum 0.9, initial learning rate 0.1, and cosine annealing. Training applies a random
 287 224×224 resized crop and horizontal flip; validation resizes to 256 pixels and takes a 224×224
 288 center crop. Both splits use standard ImageNet normalization. The highest clean validation-accuracy
 289 checkpoint is converted with $L = T = 16$ and evaluated on the 50,000-image validation split.
 290 All-parameter and weights-only L2 use coefficient 10^{-4} , MNE-L2 uses 10^{-4} , and weights-only
 291 L1 uses 10^{-5} . No coefficient sweep or clean-accuracy matching was performed for the reported
 292 ImageNet comparison, and coefficients were not selected from the noisy evaluation curves.

293 For the CIFAR arithmetic estimate, dense ANN energy and event-driven SNN energy are

$$E_{\text{ANN}} = N_{\text{MAC}} E_{\text{MAC}}, \quad E_{\text{SNN}} = T N_{\text{MAC}}^{(1)} E_{\text{MAC}} + N_{\text{AC}}^{(>1)} E_{\text{AC}}, \quad (9)$$

294 with $E_{\text{MAC}} = 4.6$ pJ and $E_{\text{AC}} = 0.9$ pJ. The first convolution receives analog pixels at every
 295 timestep; all later convolutional and linear layers use measured presynaptic events. Only clean
 296 ($\sigma = 0$) activity is used in this estimate.

297 C.1 Full CIFAR robustness and energy statistics

298 Table 3 reports the full $\sigma = 1$ timestep sweep for the main comparison. Tables 4 and 5 report
 299 clean accuracy and energy at every timestep; Table 6 adds per-sample activity distributions and the
 300 L2-wo/L1 controls at $T = 16$. The main-paper Table 1 reports the four-method $\sigma = 5$ comparison.
 301 For energy accounting, we count dense operations in the first convolution at every timestep and
 302 event-triggered ACs in all later convolutional and linear layers. Every table uses all 10,000 examples
 303 in each test set; uncertainty reflects variation across five independently trained checkpoints, not
 304 measurement noise from physical hardware.

Table 5: Full clean CIFAR VGG16 arithmetic-energy estimates (mean $\pm$ sample standard deviation across seeds 40–44). ANN energy is 1.5277 mJ on CIFAR-10 and 1.5294 mJ on CIFAR-100. “Ratio” is ANN energy divided by estimated SNN energy.

Dataset	Method	$T = 4$		$T = 8$		$T = 16$	
		Energy (mJ)	Ratio	Energy (mJ)	Ratio	Energy (mJ)	Ratio
CIFAR-10	L2-all	0.1421 $\pm$ 0.0015	10.75 $\pm$ 0.11	0.2877 $\pm$ 0.0030	5.31 $\pm$ 0.05	0.5848 $\pm$ 0.0060	2.61 $\pm$ 0.03
	L2-wo	0.1227 $\pm$ 0.0008	12.45 $\pm$ 0.08	0.2483 $\pm$ 0.0016	6.15 $\pm$ 0.04	0.5050 $\pm$ 0.0033	3.03 $\pm$ 0.02
	L1-wo	0.0975 $\pm$ 0.0010	15.67 $\pm$ 0.17	0.1969 $\pm$ 0.0022	7.76 $\pm$ 0.08	0.4019 $\pm$ 0.0046	3.80 $\pm$ 0.04
	MNE-L2 (ours)	0.1141 $\pm$ 0.0005	13.39 $\pm$ 0.06	0.2312 $\pm$ 0.0012	6.61 $\pm$ 0.03	0.4716 $\pm$ 0.0026	3.24 $\pm$ 0.02
CIFAR-100	L2-all	0.1354 $\pm$ 0.0020	11.30 $\pm$ 0.16	0.2742 $\pm$ 0.0039	5.58 $\pm$ 0.08	0.5577 $\pm$ 0.0078	2.74 $\pm$ 0.04
	L2-wo	0.1271 $\pm$ 0.0005	12.03 $\pm$ 0.04	0.2569 $\pm$ 0.0010	5.95 $\pm$ 0.02	0.5225 $\pm$ 0.0021	2.93 $\pm$ 0.01
	L1-wo	0.1043 $\pm$ 0.0011	14.66 $\pm$ 0.16	0.2109 $\pm$ 0.0025	7.25 $\pm$ 0.09	0.4294 $\pm$ 0.0052	3.56 $\pm$ 0.04
	MNE-L2 (ours)	0.1197 $\pm$ 0.0009	12.77 $\pm$ 0.10	0.2424 $\pm$ 0.0019	6.31 $\pm$ 0.05	0.4942 $\pm$ 0.0037	3.09 $\pm$ 0.02

Table 6: Clean activity distribution, clean arithmetic energy, and $\sigma = 1$ robustness at $T = 16$. Within each seed, firing density is the fraction of nonzero entries across all IF outputs and is summarized over 10,000 samples by its mean, median, and 95th percentile; entries report the mean $\pm$ sample standard deviation of these statistics across seeds 40–44. Energy uses clean mean activity only.

Dataset	Method	Clean acc. (%)	Fire mean	Fire median	Fire p95	Energy (mJ)	Acc. at $\sigma = 1$ (%)
CIFAR-10	L2-all	90.59 $\pm$ 0.13	0.105 $\pm$ 0.003	0.105 $\pm$ 0.003	0.120 $\pm$ 0.004	0.585 $\pm$ 0.006	23.30 $\pm$ 6.46
	MNE-L2 (ours)	90.38 $\pm$ 0.14	0.083 $\pm$ 0.001	0.083 $\pm$ 0.001	0.099 $\pm$ 0.002	0.472 $\pm$ 0.003	90.09 $\pm$ 0.10
	L2-wo	91.10 $\pm$ 0.07	0.102 $\pm$ 0.001	0.103 $\pm$ 0.001	0.118 $\pm$ 0.001	0.505 $\pm$ 0.003	90.82 $\pm$ 0.11
	L1-wo	89.87 $\pm$ 0.20	0.077 $\pm$ 0.001	0.077 $\pm$ 0.001	0.090 $\pm$ 0.002	0.402 $\pm$ 0.005	89.60 $\pm$ 0.16
CIFAR-100	L2-all	62.27 $\pm$ 0.37	0.106 $\pm$ 0.003	0.106 $\pm$ 0.003	0.121 $\pm$ 0.004	0.558 $\pm$ 0.008	14.64 $\pm$ 3.16
	MNE-L2 (ours)	59.62 $\pm$ 0.22	0.090 $\pm$ 0.001	0.090 $\pm$ 0.001	0.107 $\pm$ 0.002	0.494 $\pm$ 0.004	59.48 $\pm$ 0.23
	L2-wo	63.83 $\pm$ 0.26	0.105 $\pm$ 0.001	0.106 $\pm$ 0.001	0.120 $\pm$ 0.001	0.522 $\pm$ 0.002	63.79 $\pm$ 0.25
	L1-wo	53.13 $\pm$ 0.73	0.084 $\pm$ 0.002	0.085 $\pm$ 0.002	0.101 $\pm$ 0.002	0.429 $\pm$ 0.005	53.08 $\pm$ 0.76

305 C.2 Depth transfer across VGG13 and VGG19

306 We repeat the post-input-IF Gaussian-noise sweep at $L = T = 16$ using VGG13 and VGG19 on
 307 CIFAR-10 and CIFAR-100. Each curve reports the mean and sample standard deviation across
 308 independently trained seeds 40–44. In addition to the four main regularizers, this control includes
 309 Calibrated MNE with the fixed $\alpha = 0.1$ configuration; α is a prescribed training schedule rather than
 310 a gradient-triggered or adaptive choice.

311 Across VGG13/16/19, MNE-L2 attains 74.28/72.06/73.23% at $\sigma = 5$ on CIFAR-10 and
 312 44.01/41.14/37.60% on CIFAR-100. Calibrated MNE closely follows L2-wo despite matching
 313 or slightly exceeding its clean accuracy, while L1-wo incurs a substantial CIFAR-100 clean-accuracy
 314 cost that grows from VGG13 to VGG19. Thus the high-noise MNE-L2 advantage transfers across
 315 the three completed depths, although the CIFAR-100 endpoint declines with depth.

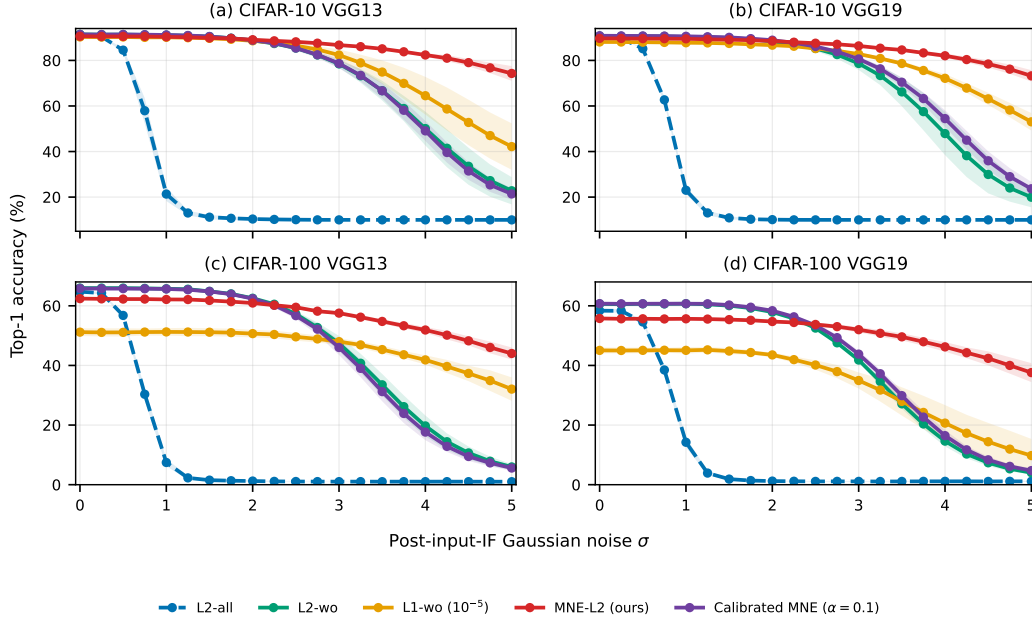


Figure 4: Depth-transfer robustness at $L = T = 16$. Gaussian noise is injected after the input IF. Curves are five-seed means and bands show sample standard deviations. MNE-L2 is highlighted in red; Calibrated MNE is included only as an appendix control.

Table 7: Additional-depth accuracy endpoints (mean $\pm$ sample standard deviation across seeds 40–44). The best value within each architecture, dataset, and noise condition is bold.

Architecture	Method	CIFAR-10		CIFAR-100	
		$\sigma = 0$	$\sigma = 5$	$\sigma = 0$	$\sigma = 5$
VGG13	L2-all	90.93 $\pm$ 0.14	9.99 $\pm$ 0.01	64.73 $\pm$ 0.19	1.01 $\pm$ 0.01
	L2-wo	91.44 $\pm$ 0.14	22.78 $\pm$ 5.99	65.94 $\pm$ 0.15	5.96 $\pm$ 0.96
	L1-wo	90.30 $\pm$ 0.14	42.17 $\pm$ 10.20	51.16 $\pm$ 1.31	32.06 $\pm$ 3.92
	MNE-L2 (ours)	90.61 $\pm$ 0.19	74.28 $\pm$ 3.25	62.41 $\pm$ 0.15	44.01 $\pm$ 2.02
	Cal. MNE	91.41 $\pm$ 0.13	21.38 $\pm$ 2.58	65.82 $\pm$ 0.22	5.59 $\pm$ 0.92
VGG19	L2-all	90.19 $\pm$ 0.12	10.00 $\pm$ 0.00	58.37 $\pm$ 0.64	1.13 $\pm$ 0.18
	L2-wo	90.69 $\pm$ 0.18	19.95 $\pm$ 4.44	60.62 $\pm$ 0.32	4.03 $\pm$ 0.65
	L1-wo	88.09 $\pm$ 0.25	53.04 $\pm$ 3.85	45.04 $\pm$ 1.01	9.77 $\pm$ 5.59
	MNE-L2 (ours)	89.67 $\pm$ 0.06	73.23 $\pm$ 2.50	55.75 $\pm$ 0.43	37.60 $\pm$ 3.19
	Cal. MNE	90.84 $\pm$ 0.03	23.66 $\pm$ 3.05	60.73 $\pm$ 0.22	4.73 $\pm$ 0.76

316 C.3 Compact MNIST-family controls

317 We use MNIST [27] and Fashion-MNIST [28] with a 2,081-parameter CNN: two 3×3 convolution-
 318 BN-IF blocks with 2 and 4 channels, 2×2 max pooling, and a linear classifier. Source ANNs are
 319 trained for 30 epochs with seeds 40–44 and converted at $T \in \{4, 8, 16\}$. We compare MNE-L2,
 320 orthogonal weight regularization, no regularization, and optimizer L2. During the low-latency sweep,
 321 Gaussian noise is added after both the input IF and first hidden IF at $\sigma \in \{0, 0.25, 0.5\}$; each nonzero
 322 point averages three paired streams within each training seed.

323 For the compact CNN, the dense ANN baseline is 30,184 MACs per sample. The event counter
 324 charges every nonzero presynaptic activation once for each valid downstream connection. It counts
 325 both convolutional layers and the classifier, including edge effects from padding, and accumulates
 326 over all T timesteps. Reported means and standard deviations are across five independently trained
 327 checkpoints.

328 Figure 5 gives the compact low-latency view. At $T = 4$, the event-SynOps ratio ranges from 0.36 to
 329 0.45 on Fashion-MNIST and 0.63 to 0.65 on MNIST; at $T = 16$, it rises to 1.54–1.97 and 2.71–2.86.
 330 These are algorithmic operation counts rather than wall-clock latency or hardware energy [1, 26].

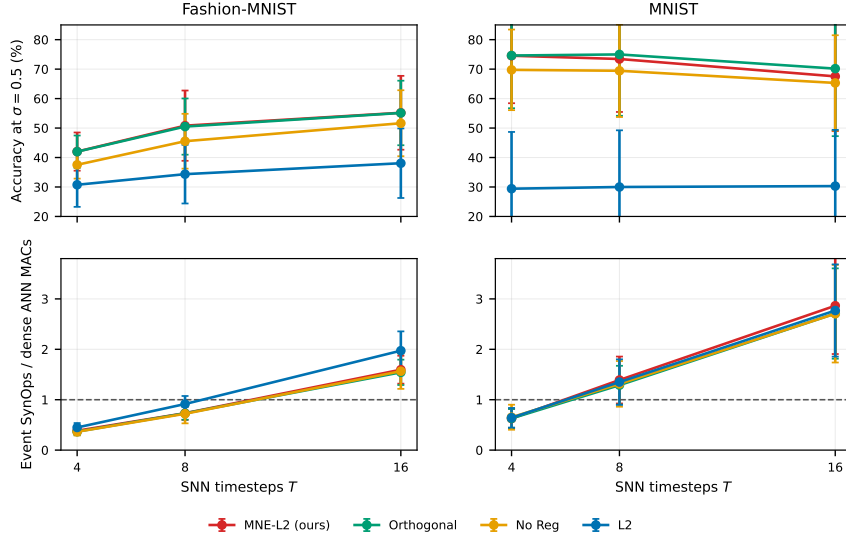


Figure 5: Appendix compact-model control. Top: five-seed accuracy under two-site Gaussian noise with $\sigma = 0.5$. Bottom: event-driven synaptic-operation proxy relative to one dense ANN MAC pass; the dashed line marks ratio 1. Error bars are standard deviations across training seeds.

Table 8: Full five-seed accuracy at $\sigma = 0.5$ (mean $\pm$ standard deviation across training seeds). Noise is injected after both the input IF and first hidden IF.

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
Fashion	MNE-L2 (ours)	42.01 ± 6.48	50.81 ± 11.93	55.19 ± 12.53
Fashion	Orthogonal	41.98 ± 5.49	50.50 ± 9.54	55.12 ± 10.95
Fashion	No Reg	37.53 ± 4.69	45.53 ± 9.30	51.64 ± 11.19
Fashion	L2	30.75 ± 7.49	34.35 ± 9.95	38.06 ± 11.80
MNIST	MNE-L2 (ours)	74.55 ± 16.12	73.44 ± 17.98	67.54 ± 18.60
MNIST	Orthogonal	74.62 ± 17.89	74.99 ± 20.82	70.17 ± 22.92
MNIST	No Reg	69.75 ± 13.68	69.44 ± 15.69	65.31 ± 16.18
MNIST	L2	29.41 ± 19.27	30.00 ± 19.23	30.30 ± 19.07

D Reproducibility Notes

All reported datasets use their standard public train/test splits. The evaluation code fixes the training seed separately from the perturbation seed, stores every per-seed result before aggregation, and reports sample standard deviation rather than standard error. Figures 1, 2, and 4 are generated directly from stored CSV summaries by a deterministic plotting script; curve values are neither redrawn nor transcribed. The compact event-count control uses CPU inference and 1,024 examples per checkpoint. The CIFAR activity evaluation uses each full 10,000-example test set: sample mean, median, and p95 are computed within each seed, then summarized across seeds 40–44 by mean and sample standard deviation. Only $\sigma = 0$ activity enters energy; $\sigma = 1$ activity is retained as a diagnostic and excluded from deployment estimates. Full VGG and ResNet training requires GPU compute; evaluation does not update model parameters.

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